will require enhanced intelligence to self configure in adhoc networks and having a routing capabilities to route over a packet-switched network.

### Fifth Generation (5G) 2019+

5G is the stage succeeds the 4G (LTE/ WiMAx), 3G(umts) and 2G(GSM) syetems. 5G targets to perform the high data rate, reduced latency, energy saving, cost reduction, higher system, capacity, and massive device connectivity. The two phases of 5G, First one will be Release-15 complete by March 2019,Second one Release-16 is expected to complete at March2020, for submission to the ITU(International Telecommunication Union) as a candidate IMT-2020 technology. The ITU IMT – 2020 provides speed up to 20 gigabits per second it has been demonstrated with millimeter waves of 15 gigahertz and higher frequency. 3 GPP standard includes any network using New Radio software. 5G New Radio can access at lower frequencies from 600 MHz to 6 GHz. Speed in the lower frequencies are only modest higher than 4G systems, estimated at 15% to 50% faster.

## 11.1.3 WLANS 802.11

Wi-Fi stands for Wireless Fidelity. It is a wireless network technology that permits computers and alternative devices to be connected to every alternative into a local area network and to the net without wires and cables. Wi-Fi is additionally stated as wireless local area networkthat stands for wireless local area network, and 802.11, is the technical code for the protocol. See Figure 11.6

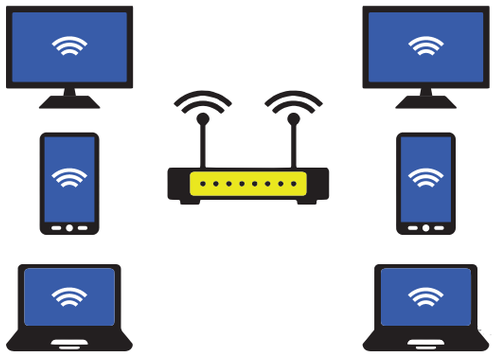


Figure 11.6 Wi-Fi

**ADVANTAGES: Benefits of Wi-Fi are**

- It provides mobility. Example: I get Internet connection wireless through my laptop computer at home and at work, because of Wi-Fi, hotspots both at home and at work can be used.
- It provides connection to Internet.
- Flexibility of LAN.
- Ensures connectivity.
- It allows remote places to benefit from connectivity.
- Low cost, high benifts.

## 11.1.4 RFID

- RFID - Radio Frequency Identification.

RFID is a technology designed to locate objects (Credit cards, Passports or even livestock) using radio signals.

RFID used radio waves to read and capture information stored on a tag attached to an object. Tag can be read from several feet away and does not need to be in direct-line-of-sight of the reader to be tracked. RFID has been made up of two parts a reader and a tag or a label. RFID tags are installed with a transmitter and receiver.

RFID component on the tags has two parts: a microchip which stores and processes the information, and the antenna



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